Top Performing Students

Given an array of students:

```
[
 { name: "John", marks: [80, 90, 85] },
 { name: "Emma", marks: [95, 92, 98] },
 { name: "Alex", marks: [60, 70, 65] }
]
```

Task

- Calculate each student's average.
- Keep only students with average ≥ 85.
- Return their names sorted by average descending.

2. E-commerce Revenue Analysis

```
[
 { category: "Electronics", amount: 500 },
 { category: "Books", amount: 100 },
 { category: "Electronics", amount: 300 },
 { category: "Clothing", amount: 200 }
]
```

Task

- Group sales by category.
- Calculate total revenue per category.
- Return categories with revenue greater than 300.

3. Most Frequent Words

Given a paragraph:

"The quick brown fox jumps over the lazy dog. The fox is quick."

Task

- Remove punctuation.
- Convert all words to lowercase.
- Count occurrences of each word.
- Return the top 3 most frequent words.

4. Employee Department Statistics

```
[
  { name: "A", dept: "IT", salary: 50000 },
  { name: "B", dept: "HR", salary: 40000 },
  { name: "C", dept: "IT", salary: 60000 }
]
```

Task

- Group employees by department.
- Calculate average salary per department.
- Return department with highest average salary.

5. Product Inventory Dashboard

```
[
  { name: "Laptop", stock: 5, price: 50000 },
  { name: "Phone", stock: 0, price: 20000 },
  { name: "Tablet", stock: 10, price: 15000 }
]
```

Task

- Remove out-of-stock products.
- Calculate inventory value (stock × price).
- Find total inventory value.

6. Social Media Engagement Analysis

```
[
  { postId: 1, likes: 100, comments: 20, shares: 10 },
  { postId: 2, likes: 200, comments: 50, shares: 30 }
]
```

Task

- Calculate engagement score:
  - likes × 1
  - comments × 2
  - shares × 3
- Return posts with engagement score > 300.

7. Movie Recommendation Engine

```
[
  { title: "Movie A", rating: 8.5, genres: ["Action"] },
  { title: "Movie B", rating: 9.0, genres: ["Drama"] }
]
```

Task

- Filter movies rating > 8.
- Extract genres.
- Count how many times each genre appears.

8. Nested Orders Analysis

```
[
  {
    customer: "John",
    orders: [
      { product: "Laptop", amount: 50000 },
      { product: "Mouse", amount: 1000 }
    ]
  }
]
```

Task

- Flatten all orders.
- Calculate total spending per customer.
- Find highest spending customer.

9. Bank Transaction Report

```
[
  { type: "credit", amount: 1000 },
  { type: "debit", amount: 300 },
  { type: "credit", amount: 500 }
]
```

Task

- Separate credits and debits.
- Calculate total credits and debits.
- Find final balance.

10. User Activity Tracking

```
[
  { user: "A", actions: ["login", "logout"] },
  { user: "B", actions: ["login", "purchase", "logout"] }
]
```

Task

- Count total occurrences of each action.
- Find most common action.

11. GitHub Repository Analytics

```
[
  { repo: "A", stars: 100, forks: 50 },
  { repo: "B", stars: 200, forks: 80 }
]
```

Task

- Calculate popularity score:
    - stars × 2 + forks
- Return top repository.